

BitSaver: Recovering BitLocker Partitions

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Abstract—With a majority of industries and homes beginning to practice data encryption, BitLocker disks are becoming more and more common in digital forensic fields. BitSaver is a new command-line interface (CLI) program that detects and finds BitLocker partitions and rebuilds partition tables to allow an encrypted disk to be readable again, assuming that a BitLocker recovery key exists and is in the possession of law enforcement or data forensic teams.

Index Terms—encryption, data, forensics, BitLocker, recovery.

I. INTRODUCTION

A. BitLocker and Digital Forensics

BitLocker is an application that is very relevant in the field of Information Technology (IT) and digital forensics. Most enterprises have policies that allow their systems to automatically and routinely encrypt data. This is both to prevent confidential data from being stolen. However, this often creates a new barrier for digital forensics teams when they need to analyze a disk for official reason such as law enforcement. According to North Forensics, encryption makes data unreadable without a key, bypassing even the most advanced forensic tools. This barrier is also time-consuming and complex for technicians to overcome, sometimes impossible.

In addition, the legal aspects of evidence that utilizes encryption can be complicated. Meticulous documentation is required to maintain integrity of an encrypted piece of evidence so that it can be used in the court of law [North Forensics].

B. BitSaver Objectives and Motivation

When an individual manually deletes a BitLocker partition, the data does not get deleted. Instead, the disk marks the partition as unallocated space. Thus, the data remains on the drive. However, the data cannot be read directly due to encryption. Suspects who are informed on disk partitioning schemes may delete a BitLocker partition in order to stall law enforcement. Standard data recovery will not work on a disk that has BitLocker as the data is unreadable prior to decryption.

The goal of this BitSaver is to create a tool that can automate the recovery of BitLocker partitions that have been turned into unallocated space on a disk. Typically, this happens when a storage device has not been used or formatted. However, it can also occur when an individual manually deletes partitions such as with Windows DiskPart. BitSaver is focused on creating a

tool that works for Windows as that is the supermajority of systems that utilizes BitLocker. Thus, BitSaver does not work for other operating systems like MacOS or Linux/UNIX.

Law enforcement, digital forensics, and enterprise IT technicians will be able to use this application to recover BitLocker partitions that have been deleted and not formatted. For example, if a suspect knows that law enforcement is on their trail, the suspect may delete BitLocker partitions and not format the deleted partitions of their devices to obscure the data but still can recover the data at a later point (post-investigation). The suspect would still need to hide their recovery keys, but a barrier to information has formed for law enforcement. Another example would be if a trainee IT technician accidentally deleted a BitLocker partition on an important, critical data drive while trying to install the Windows operating system for another partition. There is already many techniques for rewriting boot records to restore functionality. However, most tools require the user to perform potentially destructive disk writes, which can cause permanent data loss. This is where the application can be very useful. BitSaver automatically detects and locates BitLocker partitions, accurately calculates partition sizes, and writes boot records for its users.

C. BitSaver Plans and Implementation

To implement this tool, standard Python libraries, PowerShell commands, and Runtime Software's open source, free DriveDoppel executable were used. Python was used to implement a user-friendly interface to communicate with the user. Standard libraries like Hashlib, Hexdump, Ctypes, etc. were also used for their ability to analyze disk data. PowerShell was used to communicate system calls to the disk safely in Windows. Lastly, DriveDoppel was used to safely write boot records to a disk.

When ran, the program will open up an interface that allows users to select numbers for their choices. The program will collect information about all disks detected on the system, including internal and external storage devices. Afterwards, the device will allow users to perform actions on disks such as disk selection, hexdumping BitLocker sectors, and repair BitLocker boot records. The system also intentionally stops users from selecting the boot drive as an option to prevent damage on the operating system. A depiction of the menu is shown in Fig. 1. Users in this screen are able to see the hostname of the machine they are working on and see many useful attributes of the drives connected to the machine such

```

Computer Name: DESKTOP-L2Q5G34

disk_number: 0
size_bytes: 1000204886016
bus_type: NVMe
model_name: Samsung SSD 970 EVO Plus 1TB
is_system: True
is_boot: True
is_readonly: False
is_removable: False
partition_style: GPT
letters: CE
volume_type: Operating System
volume_status: Fully Decrypted
capacity_GB: 930.677734

disk_number: 1
size_bytes: 15728640512
bus_type: USB
model_name: ASolid USB
is_system: False
is_boot: False
is_readonly: False
is_removable: False
partition_style: MBR
letters:
volume_type: Unknown
volume_status: Unknown
capacity_GB: 0.0

===Main Menu===
1. List Disks
2. Select Disk
3. Exit
Enter choice: |

```

Fig. 1. BitSaver CLI for users to choose options.

as drive number, full enterprise drive name, encryption status, and capacity.

II. METHODOLOGY

A. Gathering Disk Information

A easy and reproduceable testing environment must be created. A small 16 GB USB drive was used as the form of storage device in BitSaver's tests. Due to the time it takes to encrypt and decrypt a large drive, larger drive sizes were not considered for testing. The 16 GB USB drive was formatted to FAT32 and the recovery key was stored externally for unlocking.

Disk 0 Basic 931.50 GB Online	100 MB Healthy (EFI System Partition)	(C:) 930.68 GB NTFS Healthy (Boot, Page File, Crash Dump, Basic Data Partition)
Disk 1 Removable 14.65 GB Online	(D:) 14.65 GB FAT32 (BitLocker Encrypted) Healthy (Primary Partition)	

Fig. 2. BitSaver CLI for users to choose options.

Afterwards, the disk's BitLocker partition was deleted via Windows DiskPart. This resulted in Windows Disk Manager recognizing the partition as unallocated. Accessing the data also prompts a formatting request (which should not be accepted to preserve the data). Standard data recovery software yields no results when files are analyzed for recovery such as Get Data Back Pro (GDBP).

When a partition deletion occurs, the disk's partition table has been modified to clear off the partition as allocated. A rewrite of the boot record on the disk is required to restore functionality to the disk. However, the boot record is incredibly sensitive and requires an accurate approximation of the size of the BitLocker partition to avoid deleting unintended data.

Thus, the first step required is to poll the data of all disks to gather information about the partitions on each disk. The program utilizes a Windows built-in PowerShell command Get-BitlockerVolume and Get-Disk to gather and format data regarding all data storage on the system. Some of the most important information gathered about the disks is the disk number, size, boot status, drive letters, and BitLocker status. All of this information is necessary to perform accurate disk operations.

Afterwards, it marks the disk that the program is running off as unprocessable to protect the native boot drive from destructive disk writes. If selected, the program will error out and prompt user to choose a different disk. All other disks are allowed to be operated on. If a valid, non-boot disk is selected, deeper disk data is gathered via Windows Get-Disk such as logical sector size, Logical Block Addressing (LBA) sector count, and true size. Logical sector size is the total size of a single sector measured in bytes. The LBA sector count is the number of logical sectors on the storage device. Lastly, the total size is the number of bytes that the device can hold. The LBA sector count can be calculated by:

$$s/l = \gamma \quad (1)$$

where s is the capacity of the storage device in bytes, l is the logical sector size, and γ is the LBA sector count. The LBA sector count is crucial to correct disk operations as a disk is divided and numbered by these sectors.

B. Finding BitLocker Sector and Count

C. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc,

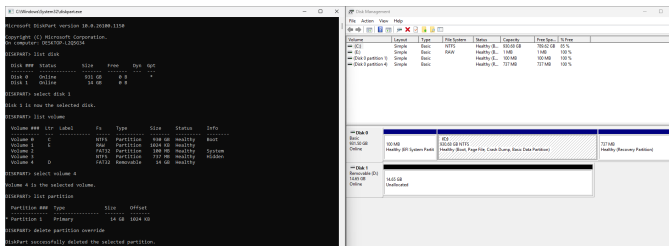


Fig. 3. BitSaver CLI for users to choose options.

and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

D. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m²” or “webers per square meter”, not “webers/m²”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm³”, not “cc”).

E. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (2)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(2)”, not “Eq. (2)” or “equation (2)”, except at the beginning of a sentence: “Equation (2) is . . .”

F. $\LaTeX$ -Specific Advice

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in $\LaTeX$ will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

$\BIBTeX$ does not work by magic. It doesn’t get the bibliographic data from thin air but from .bib files. If you use $\BIBTeX$ to produce a bibliography you must send the .bib files.

$\LaTeX$ can’t read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

$\LaTeX$ does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it’s supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

G. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

H. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

I. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

J. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 4”, even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In

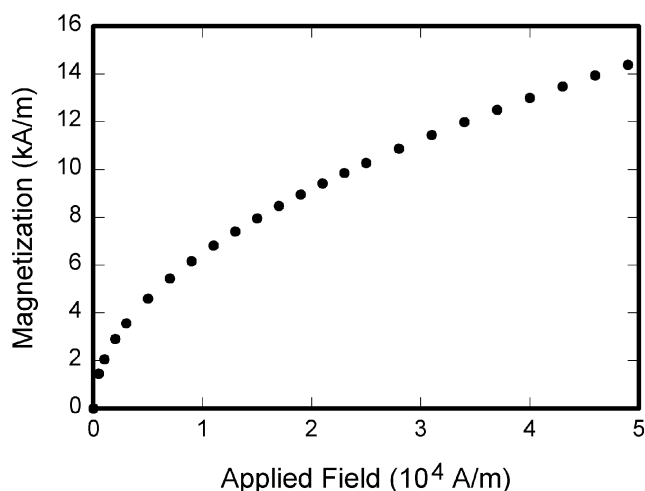


Fig. 4. Example of a figure caption.

the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first ...”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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